AI.106

Control Systems

COURSE OVERVIEW

Classical control theory: modeling dynamic systems, feedback, stability, and PID—foundations for robotic motion.



ILLUSTRATIVE HERO IMAGERY

1

SESSION

60 MINUTES

ORIENTATION: WHY CONTROL MATTERS

KEY TOPICS

- Open-loop vs. closed-loop
- Control objectives: regulate and track
- Examples in robotics and AI
- Course roadmap

TEACHING FOCUS

Motivate feedback control as the regulator of behavior.

KEY TAKEAWAY Control makes behavior stable and predictable.

2

SESSION

60 MINUTES

MODELING DYNAMIC SYSTEMS

KEY TOPICS

- Differential equations and state space
- Transfer functions and Laplace
- Mechanical and electrical examples
- Block diagrams

TEACHING FOCUS

Describe a system mathematically before controlling it.

▲ **KEY TAKEAWAY** A model is the controller's blueprint.

3

SESSION

60 MINUTES

TIME-DOMAIN ANALYSIS

KEY TOPICS

- Step response and transient specs
- Rise time, settling time, overshoot
- Steady-state error
- First- and second-order systems

TEACHING FOCUS

Predict how a system responds over time.

KEY TAKEAWAY Transient behavior reveals system character.

4

SESSION

60 MINUTES

STABILITY & FREQUENCY RESPONSE

KEY TOPICS

- Routh–Hurwitz stability
- Bode plots and gain/phase margins
- Nyquist criterion intro
- Root locus basics

TEACHING FOCUS

Guarantee a controller will not diverge.

KEY TAKEAWAY Stability is non-negotiable.

5

SESSION

⏱ 60 MINUTES

PID CONTROL

KEY TOPICS

- Proportional, integral, derivative actions
- PID tuning: Ziegler–Nichols
- PID for motor position and speed
- Anti-windup and limits

TEACHING FOCUS

Implement the most-used controller in robotics.

▲ **KEY TAKEAWAY** PID is the workhorse of motion control.

CONTROL SYSTEMS

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SESSION

60 MINUTES

CONTROL IN ROBOTICS & REVIEW

KEY TOPICS

- Cascade and feedforward control
- Digital control and sampling
- Course recap and assessment
- Toward intelligent control (AI.308)

TEACHING FOCUS

Apply control to real robotic platforms.

KEY TAKEAWAY Control turns intent into motion.